

Ashwin M R

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EDUCATION

University of Illinois at Urbana–Champaign (UIUC), USA

2024 – 2025

Master of Engineering in Aerospace Systems Engineering, Grainger College of Engineering

Research with Prof. Jason Merret on MBSE integration with Generative AI for aerospace systems

Activities: Illini Chess Club; Illini Run Club

EXPERIENCE

Indian Space Research Organization (ISRO), Thermal Simulation Lab — Trivandrum, India

Jan 2024 – Jul 2024

Graduate Apprentice

- Developed MATLAB code to convert 3D ray-tracing coordinates into ANSYS CFX input format.
- Performed ray-tracing analysis of a 500 mm filament IR lamp on a High-Performance Radiant Heater (HPRH) targeting uniform heat flux of 130 W/cm^2 over a $200 \text{ mm} \times 200 \text{ mm}$ specimen.
- Tested an 80 W CO_2 laser on a $50 \text{ mm} \times 50 \text{ mm}$ specimen using a laser power meter and thermopile sensor.
- Tested improved Parabolic Reflector Module configurations with IR lamps simulating concentrated-surface heating up to 180 W/cm^2 .

TECHNICAL PROJECTS

ThermalPath — Open-Source Thermal Design Library for Electronics Cooling

- Open-source Python library for checking thermal design calculations in electronics cooling, with typed SI-validated inputs and reproducible examples.
- Implemented steady series-resistance and conductance-matrix network solvers with connectivity and signed heat-balance diagnostics; added transient modeling — one-node RC step response, backward-Euler networks with step-energy verification, and a constant-conductivity finite-volume plate solver on a rectangular structured grid.

OpenFOAM CFD Benchmark and Thermal Simulation Suite

- Reproducible OpenFOAM 13 workflows for canonical CFD benchmarks: backward-facing step, cylinder vortex shedding, buoyant cavity, NACA 4412 trailing-edge separation, periodic hill, NASA wall-mounted hump, non-Newtonian channel flow.
- Multiphase boiling in heated microchannels (compressibleVoF, Lee-type phase change, CHT) and CHT analysis of air-cooled CPU/heat-sink assemblies.
- Audit-first validation: benchmark-mapped verification and frozen evidence artifacts separating real gains from numerical noise.

Topology-Aware MBSE Extraction from Aerospace System Descriptions

- Reproducible pipeline converting aerospace technical prose into validated MBSE models; exports UML/DI artifacts for Eclipse Papyrus.
- 100-case benchmark with paired reference models; compared spaCy extraction vs schema-constrained LLM extraction via precision/recall/F1 and graph edit distance.
- Perturbation-robustness studies and a nine-model benchmark on topology error, latency, and cost tradeoffs.

Adaptive Model Reduction for Incompressible CFD

- OpenFOAM workflow collecting full-order snapshots, forces, and validation signals; pressure-aware ROM with adaptive rollouts gated on stability, force, pressure-residual, accuracy, and runtime.
- Validated DFG 2D-2 cylinder flow against drag, lift, pressure-drop, and Strouhal targets; auditable figures and tables.

PUBLICATIONS

Journal Paper

- [Comparative analytical analysis and component selection of resistojet thruster for satellite propulsion](#), *Journal of Space Safety Engineering*, 2024.

Conference Papers

- [Structural and Thermal Analysis of a CubeSat](#), in *Lecture Notes in Mechanical Engineering*.
- [Material selection based on joule heating simulation for resistojet thruster](#), *Materials Today: Proceedings*.

SKILLS

Simulation Software: ANSYS (CFX, Fluent, Transient Structural), OpenFOAM, COMSOL Multiphysics, ABAQUS, SolidWorks, Creo Parametric, MeshLab, ParaView, MATLAB

SysML & AI Tools: SysML 2, Papyrus, NLP (spaCy), Chaos Engineering

Programming Languages: C++, Python, HTML, CSS, JavaScript

Operating Systems: Windows, Fedora, Arch Linux, Ubuntu

CFD: AMR, Multiphase Flow, Turbulence Modeling, Discretization Methods, Post-processing, CHT

Experimental Skills: TPS Characterization, Sensor Calibration and Heat Flux Mapping, IR Thermography, Radiative Property Testing, Heat Flux Gauges and Thermocouples